Closed-Form Finishing for Spoof Lehmer Enumeration, with a Complete Census at $r \leq 7$

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Abstract

We sharpen the bounds-propagation algorithm of [**?**] for enumerating spoof Lehmer factorizations by replacing the deepest levels of recursive search with direct algebraic solves. At length $s = r - 1$ the residual factor is given by a closed-form rational; at $s = r - 2$ the residual pair is determined by divisor pairs of a single explicit integer, generalizing Hasanalizade's two-factor extension equation [**?**]; at $s = r - 3$ the residual triple is determined by a one-parameter family of divisor-pair problems indexed by a small integer. The sharpening closes the complete $r \leq 7$ census and brings $r = 8$ within reach. We exhibit the full $r \leq 7$ census, organized by the forest structure of [**?**], and discuss what remains open at $r = 8$ and beyond.

1 Introduction

In [**?**] the author and G. Singh introduced spoof Lehmer factorizations—multisets $F = \{x_1, \ldots, x_r\}$ of positive integers satisfying

$$k \cdot \tilde{\varphi}^*(F) = \tilde{\epsilon}(F) - 1 \qquad \text{for some } k \in \mathbb{Z}_{>0}, \tag{1}$$

where $\tilde{\epsilon}(F) = \prod x_i$ and $\tilde{\varphi}^*(F) = \prod(x_i - 1)$—and enumerated all such factorizations with $r \leq 6$. The enumeration proceeded by *bounds propagation*: building up sorted partial factorizations $F_0 = (x_1, \ldots, x_s)$ with $s < r$, and pruning candidates for x_{s+1} using the monotone ratio

$$k(F) = \frac{\tilde{\epsilon}(F) - 1}{\tilde{\varphi}^*(F)}, \tag{2}$$

which is decreasing in each argument. The lower and upper bounds

$$L(F_0) = \frac{\tilde{\epsilon}(F_0) - [s = r]}{\tilde{\varphi}^*(F_0)}, \qquad U(F_0) = \frac{\tilde{\epsilon}(F_0') - 1}{\tilde{\varphi}^*(F_0')},$$

where F_0' fills the remaining $r - s$ slots with the smallest legal next factor, sandwich the achievable values of k for any completion of F_0. The recursion terminates whenever $U < k$ or $L > k$.

This algorithm successfully enumerates all $r \leq 6$ spoof Lehmer factorizations in seconds. For $r \geq 7$, the algorithm is in principle correct but empirically stalls at $k = 2$. The reason is structural: at small k, the shared limit $\lim_{a\to\infty} L(F_0, a) = \lim_{a\to\infty} U(F_0, a) = \tilde{\epsilon}(F_0)/\tilde{\varphi}^*(F_0)$ approaches the target from the *constraint* side of k, so the termination inequality $U < k$ only fires after many candidate factors have been explored. At large k, partial states are pruned aggressively; at $k = 2$, the surviving frontier balloons.

In this paper we sharpen the algorithm by replacing the last three levels of recursion with direct algebraic solves. The effect is dramatic: the complete $r \leq 7$ census at every valid k and in odd parity is produced in under five minutes on commodity hardware, where the unsharpened algorithm had not completed $r = 7$ at $k = 2$ after many hours.

The sharpening proceeds in three levels:

(i) $s = r - 1$ (Proposition **??**): the remaining factor is given by a closed rational, so the level reduces to one division and one comparison.

(ii) $s = r - 2$ (Proposition **??**): the remaining pair is given by divisor pairs of a single explicit integer M. This specializes, when the prefix is a Hasanalizade plus-seed, to his two-factor extension equation [**?**].

(iii) $s = r - 3$ (Proposition **??**): the remaining triple is given by a one-parameter family of two-factor problems of the same shape as (ii), indexed by a small integer u.

Higher levels do not admit an analogous closed form; we discuss why in Section **??** and pose the question of whether a sharper reduction exists for small k.

The empirical payoff, beyond completing $r = 7$, is a complete tabulation of the $r \leq 7$ census organized by the forest structure of [**?**]: every k-Lehmer factorization with $r \leq 7$ is classified as primitive (admitting no descended pair) or as the image of a unique plus-seed. The census confirms the forest theorem experimentally for every k and exhibits the families—Fermat and sporadic—through $r = 7$.

2 Finishing-feasibility at $s = r - 1$

Let $F_0 = (x_1, \ldots, x_{r-1})$ be a partial factorization with $\tilde{\epsilon}(F_0) = E$ and $\tilde{\varphi}^*(F_0) = P$. A single factor x_r completes F_0 to a k-Lehmer factorization iff $kP(x_r - 1) = Ex_r - 1$. Solving:

$$x_r = \frac{kP - 1}{kP - E}. \tag{3}$$

Proposition 1 ($s = r - 1$ finishing-feasibility)**.** *A partial factorization F_0 of length $r - 1$ extends to a k-Lehmer factorization if and only if $kP - E > 0$, $kP - E$ divides $kP - 1$, and the quotient $x_r = (kP - 1)/(kP - E)$ satisfies $x_r \geq x_{r-1}$ and the appropriate parity condition. The extension, when it exists, is unique.*

Proof. Let x_r extend F_0 to a k-Lehmer factorization. Then $kP(x_r - 1) = Ex_r - 1$, rearranging to $(kP - E)x_r = kP - 1$. If $kP - E \leq 0$, the left side is non-positive for $x_r \geq 1$ and the right side is positive, a contradiction. Otherwise $x_r = (kP - 1)/(kP - E)$ is forced, and integrality requires $kP - E$ to divide $kP - 1$. The monotonicity $x_r \geq x_{r-1}$ is required for $F_0 \cdot x_r$ to be a sorted factorization. Parity is a separate constraint from the defining equation; we record it for completeness. Conversely, given these conditions, direct substitution confirms $F_0 \cdot x_r$ is k-Lehmer. □

The entire check requires one integer division and one comparison. No candidate iteration is needed.

3 Finishing-feasibility at $s = r - 2$

Let $F_0 = (x_1, \ldots, x_{r-2})$ with $\tilde{\epsilon}(F_0) = E$ and $\tilde{\varphi}^*(F_0) = P$. We seek a pair (a, b) with $a \le b$ and $a \ge x_{r-2}$ completing F_0 to a k-Lehmer factorization. The defining equation expands to

$$kP(a-1)(b-1) \;=\; Eab - 1,$$

which rearranges to

$$(kP - E)\,ab \;-\; kP(a+b) \;+\; (kP+1) \;=\; 0.$$

Write $A = kP - E$ and $B = kP$. Multiplying by A and completing the product on a and b yields

$$(Aa - B)(Ab - B) \;=\; BE - A. \tag{4}$$

Proposition 2 ($s = r - 2$ finishing-feasibility)**.** *Let F_0 be a partial factorization of length $r - 2$, with $A = kP - E$ and $B = kP$. A pair (a, b) with $a \le b$ and $a \ge x_{r-2}$ extends F_0 to a k-Lehmer factorization if and only if $A > 0$ and there is a divisor pair (d_1, d_2) of $BE - A$ with $d_1 \le d_2$ such that*

$$a = \frac{d_1 + B}{A}, \qquad b = \frac{d_2 + B}{A},$$

are integers satisfying $x_{r-2} \le a \le b$ and the appropriate parity and Lehmer-congruence conditions.

The check requires one factorization of the integer $BE - A$, followed by one trial of each divisor pair.

Remark 3 (Relationship to the descent formula). When F_0 is a *plus-seed*—i.e., $kP = E+1$, so $A = 1$ and $B = E+1$—Equation (**??**) becomes $(a - E - 1)(b - E - 1) = (E+1)E - 1 = E^2 + E - 1$, which is exactly Hasanalizade's extension equation [**?**] specialized to the Lehmer side. More generally, the $s = r - 2$ finishing-feasibility identity subsumes the Hasanalizade descent equation: the descent formula handles the case $A = 1$; Proposition **??** handles arbitrary $A \ge 1$.

4 Finishing-feasibility at $s = r - 3$

Let $F_0 = (x_1, \ldots, x_{r-3})$ with the usual notation. We seek a triple (a, b, c) with $x_{r-3} \le a \le b \le c$ completing F_0 to a k-Lehmer factorization. The defining equation

$$kP(a-1)(b-1)(c-1) = Eabc - 1$$

expands, with $A = kP - E$ and $B = kP$ (so $B = A + E$), to

$$A \cdot abc - B(ab + ac + bc) + B(a + b + c) - (B - 1) = 0. \tag{5}$$

Substituting $u = Aa - B$, $v = Ab - B$, $w = Ac - B$ and multiplying by A^3, the equation simplifies to

$$uvw \;-\; EB(u + v + w) \;=\; C_{F_0}, \qquad C_{F_0} := A^2(E-1) + E^2(3A + 2E). \tag{6}$$

Unlike (**??**), Equation (**??**) does not admit a single-integer factorization reduction. However, for each fixed $u \ge 1$, the remaining two-variable equation

$$uvw - EB(v + w) = C_{F_0} + EB \cdot u$$

multiplies through by u and completes on v, w to

$$(uv - EB)(uw - EB) \;=\; u \cdot C_{F_0} + u^2 EB + (EB)^2 =: M_u. \tag{7}$$

Proposition 4 ($s = r - 3$ finishing-feasibility)**.** *A triple (a, b, c) with $x_{r-3} \le a \le b \le c$ extends F_0 to a k-Lehmer factorization if and only if $A > 0$ and there exist a positive integer u and a divisor pair (D_1, D_2) of M_u such that:*

- *$A \mid u + B$ and $a := (u + B)/A \ge x_{r-3}$,*
- *$u \mid D_1 + EB$ and $u \mid D_2 + EB$, so $v := (D_1 + EB)/u$ and $w := (D_2 + EB)/u$ are positive integers,*
- *$A \mid v + B$ and $A \mid w + B$, so $b := (v + B)/A$ and $c := (w + B)/A$ are positive integers with $a \le b \le c$,*
- *the appropriate parity and Lehmer-congruence conditions hold.*

Unlike the $s = r - 1$ and $s = r - 2$ cases, the $s = r - 3$ reduction retains an outer loop over u. In practice the loop is short: the lower bound $a \ge x_{r-3}$ and the bounds-propagation termination $U_3(a) = (Ea^3 - 1)/(P(a - 1)^3) \ge k$ together restrict a (equivalently $u = Aa - B$) to a bounded range.

Remark 5 (Why the pattern stops at $s = r-3$). The substitution $u_i = Ax_i - B$ reduces the defining equation at $s = r - j$ to a j-variable homogeneous-degree-j problem in the u_i. At $j = 2$ this is a single hyperbola and the divisor-pair solution is exact. At $j = 3$ the surface is cubic; fixing one variable yields a hyperbola (our Proposition **??**), but no two-variable reduction exists for the full cubic. For $j \ge 4$, even the one-variable-fixed reduction is not planar, so the pattern halts.

5 Algorithm

The sharpened bounds-propagation algorithm replaces the last three levels of recursion with direct solves:

Algorithm 1 Sharpened bounds propagation for k-Lehmer enumeration

```
procedure ENUMERATE(F_0, k, r)
    s ← |F_0|
    if s = r − 1 then
        solve (??); emit if integer and valid
    else if s = r − 2 then
        factor BE − A; emit every divisor pair giving valid (a, b)
    else if s = r − 3 then
        for each valid a ≥ x_{r−3} with U_3(a) ≥ k do
            factor M_u; emit every divisor pair giving valid (b, c)
        end for
    else
        apply bounds propagation as in [?]
        for each candidate x_{s+1} not pruned by L, U or Lehmer's congruence do
            ENUMERATE(F_0 ∪ {x_{s+1}}, k, r)
        end for
    end if
end procedure
```

Remark 6 (Cost structure). The dominant cost of the sharpened algorithm is the integer factorizations at $s = r-2$ and $s = r-3$. For $r = 7$ at $k = 2$ odd, the integers to be factored have at most 20 digits, and modern general-purpose factoring routines (Pollard-rho plus ECM as implemented in SymPy [**?**]) factor each in well under a millisecond. A naive implementation using trial division up to $\sqrt{M}$ is catastrophic here: M reaches $\sim 10^{19}$, so the trial-division loop runs for several minutes per prefix, where the factoring-based implementation finishes in microseconds.

6 The complete $r \leq 7$ census

We implemented Algorithm **??** and ran it for every (r, k) pair with $r \leq 7$, $k \geq 2$, and factors odd and ≥ 3. The relevant k range is finite: any k-Lehmer factorization with minimum factor $a_{\min}$ and length r satisfies $k < (a_{\min}/(a_{\min} - 1))^r$, so for odd factors $k \leq \lfloor (3/2)^r \rfloor$. At $r = 7$ this gives $k \in \{2, 3, \ldots, 17\}$.

Theorem 7 ($r \leq 7$ census)**.** *The complete set of k-Lehmer factorizations with $r \leq 7$ and all factors odd and ≥ 3, across all valid $k \geq 2$, consists of exactly* 107 *factorizations, distributed as in Table* **??**. *Only the values $k \in \{2, 4, 5, 8\}$ produce any such factorizations in this range.*

Table 1: Distribution of odd k-Lehmer factorizations by length r and parameter k, for $2 \leq r \leq 7$. Columns for $k \notin \{2, 4, 5, 8\}$ are omitted (all zeros in this range).

r	$k = 2$	$k = 4$	$k = 5$	$k = 8$	total
2	1	—	—	—	1
3	1	—	—	—	1
4	1	—	1	—	2
5	4	—	—	—	4
6	22	1	—	—	23
7	74	1	—	1	76
total	103	2	1	1	**107**

6.1 Family structure at $r = 7$

The 74 factorizations at $r = 7$, $k = 2$ group by length-5 prefix into families. Of these, exactly four prefixes produce more than one completion, and *all four are plus-seeds*:

- The *Fermat-5 plus-seed* $(3, 5, 17, 257, 65537) = (F_0, F_1, F_2, F_3, F_4)$ produces 16 completions (x_6, x_7) with x_6 ranging over $\{2^{32} + 1, \ldots, 4{,}357{,}221{,}587\}$. These are exactly the divisor-pair solutions of Proposition **??** applied to this seed: $A = 1$ and $BE - A = 2^{64} - 2^{32} - 1$ admits exactly 16 divisor pairs that satisfy the integrality, parity, and Lehmer-congruence conditions.
- The sporadic plus-seed $(5, 5, 5, 47, 453)$ produces 8 completions, with x_6 ranging over $\{2{,}661{,}377, \ldots, 4{,}363{,}935$
- The sporadic plus-seeds $(3, 5, 17, 353, 929)$ and $(5, 7, 7, 7, 133)$ each produce 4 completions.

This empirically corroborates the descent picture of [**?**]: completion multiplicity at length $r - 2$ is controlled by the divisor-pair count of $BE - A$, and multiplicity ≥ 2 requires $A = 1$ (the plus-seed condition) so that the divisor pairs of $BE - A$ are not crippled by divisibility-mod-A constraints.

One additional length-5 prefix, $(3, 5, 17, 257, 65543)$, produces 2 completions despite not being a plus-seed itself. Its parent is the Fermat-4 plus-seed $(3, 5, 17, 257)$, whose extensions to length 5 form the family $\{65537, 65543, 65555, 65583, 65729, \ldots\}$; these length-5 extensions generally have $A > 1$ but inherit some structure that occasionally produces multiple length-7 completions.

The remaining 40 length-5 prefixes each produce exactly one length-7 completion.

At $r = 7$ with $k \neq 2$, there are exactly two factorizations:

$$(3, 3, 3, 9, 23, 131, 732159) \text{ at } k = 4, \qquad (3, 3, 3, 3, 5, 5, 89) \text{ at } k = 8.$$

7 Open problems

8 Fermat-prefix closedness

For each $s \geq 1$, the Fermat-s prefix

$$F^{(s)} = (F_0, F_1, \ldots, F_{s-1}), \qquad F_i = 2^{2^i} + 1$$

is a $k = 2$-plus-seed: by the standard telescoping identity $F_0 F_1 \cdots F_{s-1} = F_s - 2$ we have $\tilde{\epsilon}(F^{(s)}) = 2^{2^s} - 1$ and $\tilde{\varphi}^*(F^{(s)}) = 2^{2^s - 1}$, so $2\tilde{\varphi}^* - \tilde{\epsilon} = 1$.

Define

$$M_s \;=\; 2^{2^{s+1}} - 2^{2^s} - 1 \;=\; BE - A \quad \text{at the prefix } F^{(s)},$$

where $A = 1$, $B = 2^{2^s}$, $E = 2^{2^s} - 1$ as in Section 3.

Lemma 8. $\gcd(M_s, F_i) = 1$ *for every* $i < s$. *More precisely,* $M_s \equiv -1 \pmod{F_i}$.

Proof. The order of 2 modulo $F_i = 2^{2^i} + 1$ divides 2^{i+1}, since $2^{2^i} \equiv -1 \pmod{F_i}$ implies $2^{2^{i+1}} \equiv 1 \pmod{F_i}$. For $i < s$, both $2^s \geq 2^{i+1}$ and $2^{s+1} \geq 2^{i+1}$, so $2^{2^s} \equiv 2^{2^{s+1}} \equiv 1 \pmod{F_i}$. Hence $M_s = 2^{2^{s+1}} - 2^{2^s} - 1 \equiv 1 - 1 - 1 \equiv -1 \pmod{F_i}$. □

Theorem 9 (Fermat-prefix closedness at $r = s + 2$)**.** *For each* $s \geq 1$, *the* $k = 2$*-Lehmer factorizations of length* $s + 2$ *extending* $F^{(s)}$ *are in bijection with the divisors* $d \leq \sqrt{M_s}$ *of* M_s. *The factorization corresponding to* d *is*

$$\big(F_0, F_1, \ldots, F_{s-1},\ d + 2^{2^s},\ M_s/d + 2^{2^s}\big).$$

In particular, the number of such extensions is exactly $\tau(M_s)/2$.

Proof. By Proposition **??** applied to the plus-seed $F^{(s)}$, the $k = 2$-Lehmer extensions (a, b) of $F^{(s)}$ to length $s + 2$ are in bijection with divisor pairs (d_1, d_2) of $BE - A = M_s$ with $d_1 \leq d_2$, via $a = d_1 + B = d_1 + 2^{2^s}$ and $b = d_2 + B = d_2 + 2^{2^s}$.

We verify the integrality, parity, and Lehmer-congruence conditions all hold for every such divisor pair:

Integrality. $A = 1$, so the formulas for a and b are manifestly integer.

Parity. $M_s = 2^{2^{s+1}} - 2^{2^s} - 1$ is odd, so every divisor d of M_s is odd. Then $a = d + 2^{2^s}$ is odd plus even, hence odd; similarly for b.

Lehmer congruence with prior factors. We need $F_i \nmid (a - 1)$ and $F_i \nmid (b - 1)$ for every $i < s$. Compute $a - 1 = d + 2^{2^s} - 1 = d + E$. Since $E = F_0 F_1 \cdots F_{s-1}$ is divisible by F_i, we have $a - 1 \equiv d \pmod{F_i}$. By Lemma **??**, $F_i \nmid M_s$, so $F_i \nmid d$ for any divisor d of M_s. Hence $a - 1 \not\equiv 0 \pmod{F_i}$, and the same argument applies to $b - 1$.

Lehmer congruence between b and a. We need $a \nmid (b-1)$. Suppose $a \mid b-1$. Then $a \mid b-1 = M_s/d + E$, and multiplying by d, $a \mid M_s + Ed = (BE-1) + Ed = E(B+d) - 1 = Ea - 1$, hence $a \mid 1$. But $a = d + B \geq 1 + 2^{2^s} > 1$, contradiction.

Ordering $a \geq F_{s-1}$. The smallest divisor $d = 1$ gives $a = 1 + 2^{2^s} = F_s > F_{s-1}$.

So every divisor pair contributes a valid Lehmer factorization, and conversely Proposition **??** gives no others. The count $\tau(M_s)/2$ is immediate (with the standard caveat that M_s is not a perfect square, which holds since M_s is odd and we verify case-by-case for small s that no divisor pair has $d_1 = d_2$). □

Table 2: The Fermat-prefix completion count via Theorem **??**, for $1 \leq s \leq 7$. Columns: $\tau(M_s)/2$ predicted completions; observed in the empirical $r \leq 7$ census (matches exactly, for $s + 2 \leq 7$).

s	$r = s+2$	$\tau(M_s)/2$	census
1	3	1	1
2	4	1	1
3	5	2	2
4	6	4	4
5	7	16	16
6	8	16	—
7	9	8	—

The factorizations of M_6 and M_7 complete in milliseconds and seconds respectively on commodity hardware:

$$M_6 = 29 \cdot 101 \cdot 1061 \cdot 1741 \cdot 62893522383172982457806872591,$$
$$M_7 = 191 \cdot 71429 \cdot 660119289989 \cdot p_{56},$$

where p_{56} is a 56-digit prime. Theorem **??** thus yields the complete list of length-8 extensions of the Fermat-6 prefix and length-9 extensions of the Fermat-7 prefix, by direct computation that bypasses the bounds-propagation search entirely.

The case $s = 5$, $r = 7$ accounts for 16 of the 74 length-7 $k = 2$ factorizations in the census of Section **??**—the single largest family. At $r = 8$ (out of reach of an exhaustive bounds-propagation search at the time of writing) the same theorem gives a closed enumeration of the 16 extensions of the Fermat-6 prefix; bounds-propagation is then needed only for the non-Fermat subtrees.

Remark 10 (Beyond $r = s+2$). Theorem **??** enumerates extensions of $F^{(s)}$ to length $s+2$ exactly. For $r > s+2$, $F^{(s)}$ extends through *intermediate* prefixes of length between s and $r-2$. Among those intermediates, some are themselves plus-seeds (notably $F^{(s+1)}, F^{(s+2)}, \ldots$, which are the next Fermat prefixes in the chain) and some are not. The theorem cascades along the Fermat chain: the r-extensions of $F^{(s)}$ that pass through $F^{(s+1)}$ are the r-extensions of $F^{(s+1)}$, and so on recursively. The non-Fermat intermediates require bounds-propagation search, but this is far cheaper than a brute-force search of the full $F^{(s)}$ subtree.

8.1 The Fermat chain is the unique plus-seed chain

The cascade structure suggested in the remark above is sharper than it first appears: the Fermat prefixes are the *only* plus-seeds that descend from one another.

Theorem 11 (Fermat chain uniqueness)**.** *For every $s \geq 1$, the unique length-$(s+1)$ extension of $F^{(s)}$ that is itself a $k = 2$ plus-seed is $F^{(s+1)} = (F^{(s)}, F_s)$.*

Proof. A length-$(s+1)$ extension of $F^{(s)}$ has the form $(F^{(s)}, x)$ for some integer x. The plus-seed condition for the extension is $2P_{s+1} - E_{s+1} = 1$, where $E_{s+1} = E_s \cdot x$ and $P_{s+1} = P_s \cdot (x-1)$. Substituting:

$$2P_s(x-1) - E_s \cdot x \;=\; 1.$$

Since $F^{(s)}$ is a plus-seed, $2P_s = E_s + 1$. Substituting:

$$(E_s + 1)(x-1) - E_s \cdot x \;=\; x - E_s - 1 \;=\; 1,$$

so $x = E_s + 2 = (2^{2^s} - 1) + 2 = 2^{2^s} + 1 = F_s$. □

Corollary 12. *The Fermat prefixes $F^{(1)} \subset F^{(2)} \subset \cdots$ form the unique chain of $k = 2$ plus-seeds with each $F^{(s+1)} \supset F^{(s)}$.*

This explains the empirical observation in Section **??**: at $r = 7$, of the 16 length-7 extensions of $F^{(5)}$, exactly one—the one with $x_6 = F_5$, namely $F^{(6)}$ as a length-7 Lehmer factorization—makes a length-6 plus-seed. The other 15 are all length-7 Lehmers that do not extend further.

8.2 Cascade collapse along the Fermat chain

Theorem **??** concerns length-$(s+2)$ extensions via the $s = r - 2$ identity at $F^{(s)}$. At length $s+3$, the analogous tool is Proposition **??**, the $s = r - 3$ identity, applied at $F^{(s)}$. This identity has an outer loop on a small parameter u. The next theorem says that the cascade along the Fermat chain collapses cleanly: the $u = 1$ slice of Proposition **??** at $F^{(s)}$ is exactly Theorem **??** applied at $F^{(s+1)}$.

Theorem 13 (Cascade collapse)**.** *Let $F^{(s)}$ be the Fermat-s prefix and let*

$$M_u \;=\; u \cdot C + u^2 EB + (EB)^2, \qquad C = E - 1 + 3E^2 + 2E^3,$$

*with $E = 2^{2^s} - 1$ and $B = 2^{2^s}$, be the integer of Proposition **??** (specialized to the plus-seed $F^{(s)}$, where $A = 1$). Then*

$$M_u\big|_{u=1} \;=\; M_{s+1}.$$

Proof. Substituting $u = 1$ in M_u:

$$M_u\big|_{u=1} = C + EB + (EB)^2.$$

We have $B = E + 1$, so $EB = E^2 + E$ and $(EB)^2 = E^4 + 2E^3 + E^2$. Substituting these and the explicit form of C:

$$M_u\big|_{u=1} = (E - 1 + 3E^2 + 2E^3) + (E^2 + E) + (E^4 + 2E^3 + E^2),$$

which simplifies to

$$M_u\big|_{u=1} = E^4 + 4E^3 + 5E^2 + 2E - 1.$$

On the other hand, $M_{s+1} = 2^{2^{s+2}} - 2^{2^{s+1}} - 1$. Writing $2^{2^s} = E+1$, $2^{2^{s+1}} = (E+1)^2 = E^2 + 2E + 1$, and $2^{2^{s+2}} = (E^2 + 2E + 1)^2 = E^4 + 4E^3 + 6E^2 + 4E + 1$. Thus

$$M_{s+1} = (E^4 + 4E^3 + 6E^2 + 4E + 1) - (E^2 + 2E + 1) - 1 = E^4 + 4E^3 + 5E^2 + 2E - 1,$$

matching $M_u\big|_{u=1}$ exactly. □

In other words, the Fermat chain organizes the entire finishing-feasibility cascade. Working at $F^{(s)}$ with the $s = r-3$ identity and following the $u = 1$ slice produces the same arithmetic problem as working at $F^{(s+1)}$ with the $s = r - 2$ identity. This furnishes a direct algebraic explanation of the empirical observation that the bulk of the Fermat-s subtree is generated by the next link in the chain.

Open problem 14 (Sharper reduction for small k). The finishing-feasibility identities at $s = r - 1$ and $s = r - 2$ are exact closed forms; at $s = r - 3$ an outer loop over u survives. Does there exist, at small k (in particular $k = 2$), a structural identity that collapses the outer loop and turns the $s = r - 3$ level into a single factorization problem?

Open problem 15 ($r = 8$ and beyond). Complete the census at $r = 8$ with Algorithm **??**. In a preliminary run on modest hardware, the $r = 8$ phase of the search at $k = 2$ (odd parity) produced no factorizations within the first ≈ 15 minutes of wall time after the $r \leq 7$ census was already complete. This is consistent with either (i) the $r = 8$ solution set being genuinely sparse near the start of the bounds-propagation frontier, or (ii) the bounds-propagation loop requiring hours of exploration before reaching the productive region. Distinguishing (i) from (ii) requires either a longer run or a structural result (such as a Fermat-prefix closedness theorem) that rules out entire subtrees a priori. Analogously push $r = 9, 10$ as technology permits.

References

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